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Behavioral changes and delayed colorectal cancer diagnosis during COVID-19: a real-world survey analysis

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Abstract

Background The coronavirus disease 2019 (COVID-19) pandemic profoundly altered healthcare systems and patient behavior worldwide, resulting in a significant decrease in hospital visits and delays in diagnosis and treatment. Thus, we aimed to investigate behavioral changes in patients undergoing colorectal cancer surgery during the pandemic, identify vulnerable populations, and determine the impact of such changes on cancer progression.

Methods We conducted a comprehensive survey of patients who underwent colorectal cancer surgery at Osaka University Hospital and Minoh City Hospital between January 2021 and June 2023. Participants were asked about their clinical symptoms, regular and additional medical visits, and screenings. Behavioral changes were defined as pandemic-related postponements or cancellations of additional screenings and regular medical visits. Clinical and pathological data were collected retrospectively from medical records.

Results Among the 98 patients who completed the survey, 83 met the inclusion criteria (43 males, average age 70 years). Behavioral changes due to the pandemic were observed in 8.4% of patients. Clinical symptoms, particularly rectal bleeding, were more common in the latter part of the study period ($P=0.014$). Women were more likely to change their behavior ($P=0.038$), and the disease stage tended to be more advanced in patients who changed their behavior ($P=0.062$).

Conclusions The COVID-19 pandemic led to significant behavioral changes, resulting in delayed medical consultations and more advanced cancer diagnoses. These findings highlight the need for robust public health strategies to promote regular medical checkups and timely screening, especially among vulnerable groups, during global health crises.

Keywords COVID-19 pandemic, Behavioral change, Colorectal cancer, Diagnosis delay, Patient survey

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Background

The coronavirus disease-19 (COVID-19) pandemic changed human behavior worldwide, profoundly affecting healthcare systems and patient behavior [1]. A significant decrease in hospital visits, including emergency visits, was reported during this pandemic [2]. This reduction in healthcare services also led to delayed diagnoses and treatments [3]. The number of cancer screenings dropped significantly during the pandemic [4], potentially increasing cancer morbidity and mortality for years to come. Several efforts were made to support cancer surgery services [5, 6] to avoid delayed diagnoses. However, their usefulness is challenging to determine. Nevertheless, several studies have examined the factors associated with delays in cancer treatment. A large oncology data registry in the United States revealed that Black and Hispanic patients, as well as those with metastatic disease and COVID-19 complications, experienced significant treatment delays [7]. Despite the existence of suspected causes, the reasons why specific populations were affected remain unclear.

This study aimed to fill this gap by investigating the behavioral changes experienced by patients who underwent colorectal cancer surgery during the pandemic. We conducted a comprehensive survey of patients who underwent colorectal cancer surgery to determine whether they experienced significant behavioral changes in healthcare visits during the COVID-19 pandemic. The rate of healthcare visits in Japan is significantly higher than in other countries [8]. Japanese people visit medical facilities approximately 13 times per year on average, which is one of the highest rates worldwide. In comparison, the average number of visits per year in the United States is approximately four, while in European Union countries, it ranges from six to eight. In addition, Japan has a law requiring employers to conduct annual health screenings for their employees, which often includes cancer screening [9]. This results in a higher overall participation rate in health screenings than in other countries. We aimed to assess whether behavioral changes contributed to the observed increase in advanced-stage cancers and to determine which populations were affected in Japan. By analyzing survey data, we aimed to elucidate the direct impact of these behavioral changes on cancer progression and delays in diagnosis. Additionally, this study aimed to explore strategies for effective interventions during future pandemics to prevent similar adverse outcomes. By identifying key factors that influenced patient behavior during the COVID-19 pandemic and led to delayed diagnoses, we aim to provide insights into how healthcare systems can better manage patient care during global health crises. The results of this study will contribute to the development of targeted interventions aimed at mitigating the negative impacts of delayed medical consultations and ensuring timely cancer diagnoses during future pandemics.

Materials and methods

Patients and dataset

The study included patients over 18 years of age who underwent initial treatment for colorectal cancer at either Osaka University Hospital (Osaka, Japan) or Minoh City Hospital (Minoh, Japan) between January 2021 and June 2023. The World Health Organization declared the end of the pandemic in May 2023, and surgery was typically performed within one month of cancer diagnosis. Between August 2022 and December 2023, questionnaires were distributed in person to 312 eligible outpatients who visited either institution. Participation in the survey was voluntary, and the survey was conducted using the REDCap online survey system. The inclusion criteria were completing the survey and undergoing initial treatment for a newly diagnosed case of colorectal cancer. The exclusion criteria were unmatched medical records, patient surveys, recurrence, and non-cancerous status according to the final histological examination. Patients were asked about their clinical symptoms, such as rectal bleeding, abdominal pain, and defecation abnormalities, as well as their regular and additional medical visits and screenings. All the questions are included in Supplementary Appendix 1. Additional medical screening was defined as voluntary screenings beyond regular checkups for preexisting conditions, including medical examinations by companies or individuals and cancer screenings recommended by governments or societies. Regular medical reviews were defined as routine checkups by general practitioners or specialists for preexisting conditions. Behavioral changes due to the pandemic were defined as the postponement or discontinuation of additional medical screenings and the postponement or cancellation of regular medical reviews. Clinicopathological data applicable to patients who participated in the survey were retrospectively collected from medical records. Clinical and pathological tumor stages were determined according to the National Comprehensive Cancer Network clinical guidelines [10]. This study was performed in accordance with the Declaration of Helsinki and approved by the Osaka University Ethics Committee (approval no. 20527–2) and the Minoh City Hospital Ethics Committee (approval no. R304B07). Informed consent was obtained from all patients using the opt-out method for the use of data for research purposes.

Statistical analysis

Categorical variables were analyzed using the χ^2 test, and continuous variables were analyzed using the Mann–Whitney U test. $P < 0.05$ was considered to indicate statistically significant differences. All statistical analyses

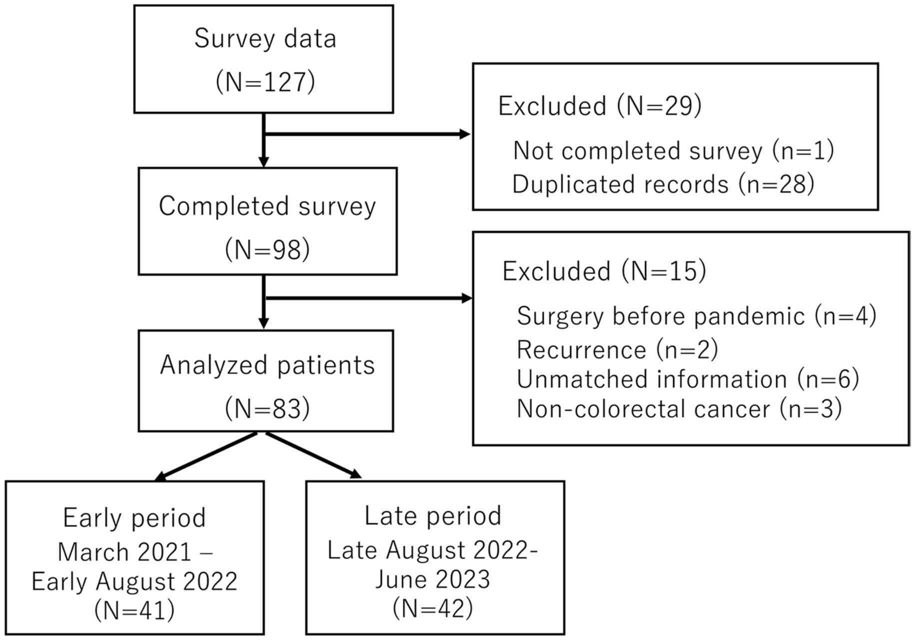

Fig. 1 Overview of the analyzed dataset

Table 1 Patient characteristics

Factors	***n* = 83**
Sex (male/female)	43/40
Age	70 (39–86)*
Operation year (2021/2022/2023)	5/59/19
Clinical symptoms (present/absent)	46/37
Rectal bleeding (present/absent)	29/54
Additional medical screening before the pandemic (present/absent/unknown)	46/35/2
Regular medical review for pre-existing conditions (present/absent/unknown)	57/24/2
Postponed or canceled additional medical screening due to the pandemic	5 (6.0%)
Postponed or canceled regular medical reviews due to the pandemic	5 (6.0%)
Behavior changes due to pandemic	7 (8.4%)**
cStage (0/I/II/III/IV)***	1/36/16/18/12
pStage (0/I/II/III/IV)***	2/31/19/21/10

*Median (range)

**Some patients reported both postponed screenings and postponed regular reviews; therefore, the total number of patients with behavioral change was seven

*** NCCN guidelines

were performed using the JMP Pro 17.1.0 statistical software program (SAS Institute, Inc., Cary, NC).

Results

From 312 eligible patients, a total of 127 surveys were returned, 29 of which were excluded due to missing information or suspected duplicate records. Consequently, 98 valid responses (31%) were obtained. Fifteen patients were excluded based on the criteria, and the responses of 83 patients were analyzed (Fig. 1). Forty-three patients were male, with a median age of 70 years (39–86). Forty-six patients underwent additional medical screenings before the COVID-19 pandemic, and 57 patients underwent regular medical reviews. Five patients postponed or canceled additional medical screenings and five postponed or canceled regular medical reviews owing to the pandemic. Because some patients reported both types of postponement, the total number of unique patients who exhibited pandemic-related behavioral changes was seven (8.4%). Patient background data are summarized in Table 1.

Patients who underwent surgery between March 2021 and early August 2022 were included in the early period ($n = 41$), and those who underwent surgery between late August 2022 and June 2023 were included in the late period ($n = 42$). No differences in sex, age, or behavioral changes existed between groups. Clinical symptoms tended to be more frequent in the late period ($P = 0.062$), as was rectal bleeding ($P = 0.014$). The clinical stage tended to be more advanced in the early period ($P = 0.056$; Table 2).

A comparison between patients with unchanged and changed behavior is shown in Table 3. Female sex was significantly associated with behavioral change ($P = 0.038$). No significant differences were observed in clinical symptoms, clinical stage, or comorbidities between the groups, although the pathological stage tended to be more advanced in patients with behavioral change ($P = 0.062$). Detailed comorbidity data are presented in Supplementary Table 1.

Table 2 Patient backgrounds over different periods

Factors	Early period (*n* = 41)	Late period (*n* = 42)	*P*-value
Sex (male/female)	20/21	23/19	0.586
Age	68 (39–86)*	73 (41–85)*	0.781
Clinical symptoms (present/absent)	19/22	28/14	<u>0.062</u>
Rectal bleeding (present/absent)	9/32	20/22	<u>**0.014**</u>
Additional medical screening before the pandemic (present/absent)	20/21	26/14	0.141
Regular medical review for pre-existing conditions (present/absent)	30/11	27/13	0.576
Postponed or canceled additional medical screening due to the pandemic (yes/no)	1/40	4/38	0.175
Postponed or canceled regular medical review due to the pandemic (yes/no)	2/28	3/24	0.554
Behavior changes due to pandemic (yes/no)	2/39	5/37	0.249
cStage (0, I, II/III, IV)**	22/19	31/11	<u>0.056</u>
pStage (0, I, II/III, IV)**	24/17	27/15	0.591

P-value < 0.1 is underlined, and P value < 0.05 is bold

*Median (range)

** NCCN guidelines

Table 3 Comparison between behavior unchanged and changed patients

Factors	Unchanged (*n* = 76)	Changed (*n* = 7)	*P*-value
Sex (male/female)	42/34	1/6	<u>**0.038**</u>
Age*	64 (50–76)*	71 (39–86)*	0.181
Clinical symptoms (present/absent)	26/50	3/4	0.646
Rectal bleeding (present/absent)	26/50	3/4	0.646
cStage (0, I, II/III, IV)**	47/29	6/1	0.141
pStage (0, I, II/III, IV)**	49/27	2/5	<u>0.062</u>
Comorbidity (present/absent)	43/33	2/5	0.155

P-value < 0.1 is underlined, and *P*-value < 0.05 is bold

*Median (range)

** NCCN guidelines

Discussion

Throughout this comprehensive study, we observed significant behavioral changes using real-world data. Although the proportion of women in this study was slightly higher than that of women with colorectal cancer in the general population, the findings are consistent with previous reports indicating a higher response rate to health questionnaires among women [11]. Therefore, the population surveyed was generally representative of the demographic backgrounds of patients with colorectal cancer in Japan [12]. Japan's universal health coverage and relatively low out-of-pocket costs have traditionally encouraged frequent medical visits and easy access to care. Even individuals with minor health issues often seek medical attention, reflecting a healthcare culture supported by institutional accessibility [13]. Therefore, the behavioral changes observed in this study represent a notable departure from Japan's usual healthcare-seeking patterns, highlighting the exceptional impact of the COVID-19 pandemic on patient behavior.

In our investigation, we analyzed two time periods during the COVID-19 pandemic, focusing on changes before and after the introduction of therapeutic drugs in Japan, particularly around September 2022. Specifically, we examined whether more patients delayed diagnosis or returned for medical consultation during the later period. We discovered that, despite the absence of significant differences in pathological stages, clinical symptoms, including rectal bleeding, were more prevalent in the later period, possibly due to delays in seeking medical care. Although the pandemic has ended, the consequences of behavioral changes are predicted to take time to resolve. Statistical analysis suggested that delayed diagnosis might persist due to uneven recovery from screening delays [14] and that treatment delays caused by the COVID-19 pandemic could increase colorectal cancer mortality rates in the long term [15]. Previous studies from various regions have similarly reported substantial disruptions in cancer care during the COVID-19 pandemic. Cancer patients experienced delays in diagnosis and interruptions in treatment, accompanied by anxiety and uncertainty regarding healthcare access [16]. A global systematic review comprehensively documented delays and disruptions in cancer care across 62 studies, most of which were driven by provider- or system-related factors such as reduced service availability, treatment postponements, and cancellations of outpatient visits. These widespread interruptions, affecting up to 77% of cancer facilities and 79% of supply chains, underscore the profound impact of the pandemic on cancer management worldwide [17]. In addition, a large national survey in the United States found that approximately one-third of patients receiving active cancer treatment or follow-up experienced care disruptions, with younger and female patients being significantly more affected [18]. These findings indicate that demographic factors, particularly sex and age, influenced healthcare-seeking behavior during the pandemic. Consistent with our results, female sex was associated with behavioral change in our cohort. From a behavioral science perspective, fear and uncertainty have been shown to influence hospital visit, while patient-reported outcomes have provided valuable insights into the psychological burden and care disruptions experienced during the pandemic [16, 19]. Collectively, these findings align with the present study, suggesting that patient-level psychological responses and risk perceptions contributed to delayed consultations and potentially more advanced disease presentations during the COVID-19 pandemic.

Additionally, we found that a subset of patients (8.4%) postponed or canceled their regular medical checkups and additional screenings because of the pandemic. This behavioral change was significant as it likely contributed to the increased number of advanced colorectal cancer diagnoses observed during the pandemic. Our study showed that the pathological stage was more advanced in patients who changed their behavior. This suggests that delays in medical consultations and screenings due to pandemic-related behavioral changes can lead to worse clinical outcomes. Although our study identified treatment delays during the pandemic, we did not assess long-term oncological outcomes. Previous reports have suggested that short delays in cancer treatment may not necessarily lead to worse prognosis [20]. However, prolonged deferrals could potentially affect disease progression, and future longitudinal analyses will be required to confirm this. Additionally, women in our study were more likely to have changed their health-seeking behaviors. Previous reports have demonstrated that women are more likely to perceive COVID-19 as a serious health problem, support restrictive public policies, and comply with them [21, 22]. These findings suggest that, in the event of a similar future pandemic, special efforts should be made to encourage women to undergo medical screenings. For future pandemic interventions, countries must thoroughly understand their societies and vulnerable groups because the more complex a society is, the more complex the selection of affected groups becomes [22, 23].

As more than half of the patients in our study had no clinical symptoms, these findings underscore the critical importance of maintaining regular medical checkups and consultations, even during global health crises. Health systems should develop robust strategies to encourage patients to continue seeking medical care and undergoing screening to avoid delays in diagnosis and treatment. Future interventions should prioritize the provision of clear information about safety measures in place at health facilities and the importance of seeking timely medical care, especially for vulnerable groups whose behaviors are affected by the pandemic.

The limitations of this study include the relatively small sample size and its conduct within a single prefecture, which may limit generalizability. However, the demographic characteristics of participants closely resembled those of patients with colorectal cancer in Japan, suggesting reasonable representativeness. The survey was administered via the REDCap electronic data capture system, with which many patients were unfamiliar. Because a paper-based version was not available, participation may have been lower compared with face-to-face or printed questionnaires. Nevertheless, considering the constraints imposed by the COVID-19 pandemic, the use of an online system offered practical advantages over mailed or printed formats.The overall valid response rate of approximately 30% was modest but acceptable for a voluntary, non-incentivized medical survey conducted during a global health crisis. Because respondents were likely more health-conscious than non-respondents, the true prevalence of behavioral changes may have been underestimated, as individuals who delayed medical care were probably less likely to participate. Only 8.4% of respondents reported behavioral changes, yet this subgroup tended to present with more advanced pathological stages, suggesting that even a limited degree of behavioral change could have meaningful clinical consequences. The observed difference in stage should be interpreted with caution but warrants confirmation in larger, multi-institutional studies. While gender differences were identified, additional factors such as socioeconomic status, employment type, and healthcare accessibility should be explored in future research. Furthermore, this study did not assess psychological factors, such as pre-existing anxiety or depression, which may have influenced behavioral changes during the pandemic. The actual duration of treatment delay could not be quantified, as the survey lacked detailed time intervals between symptom onset, diagnosis, and surgery. Future studies incorporating psychological assessment and temporal data collection are warranted to better understand the underlying mechanisms. Despite these limitations, the findings suggest that behavioral changes—though observed in a relatively small subset of patients—were not negligible and may have contributed to the higher proportion of advanced cancer at diagnosis.

In conclusion, the COVID-19 pandemic highlights the critical need for adaptive health systems to manage patient care during crises. By understanding and addressing the behavioral changes that lead to delays in cancer diagnosis and treatment, we can improve outcomes and mitigate the long-term impact of such global health events. This study provides valuable insights into patient behavior during the pandemic and suggests the need for targeted interventions for the female population in future health crises.

Abbreviation

COVID-19	Coronavirus disease 2019

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Not applicable.

Use of artificial intelligence tools
Not applicable.

Authors' contributions
CSGOCG conceptualized the study. SF and NM organized the methodology and conducted the formal analysis. RH and SK curated the data. TT, KD, MT, YS, TH, AH, and TO collected the data. MU, HY, KM, YD, and HE supervised the research. SF wrote the draft, and all authors reviewed it.

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Data availability
All data generated or analyzed during this study are included in this published article.

Declarations

Ethics approval and consent to participate
This study was performed in accordance with the Declaration of Helsinki and approved by the Osaka University Ethics Committee (approval no. 20527–2) and the Minoh City Hospital Ethics Committee (approval no. R304B07). Informed consent was obtained from all patients using the opt-out method for the use of data for research purposes.

Consent for publication
Consent for publication was obtained from all patients using the opt-out method for the use of data for research purposes.

Competing interests
The authors declare no competing interests.

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